

MAIAR GHODBANE

Master's Student in Biotechnology & Molecular Engineering

Structural Biology • Molecular Modeling • Protein Engineering • Drug Discovery

✉ maiarghodbane@gmail.com

☎ +33 6 02 73 46 10

📍 Nancy, France

Profile

Master's student in Biotechnology Molecular Engineering with a multidisciplinary background in biotechnology engineering and pharmaceutical biotechnology. Passionate about integrating experimental biotechnology, structural biology and molecular modeling to investigate biomolecular mechanisms and accelerate drug discovery. Experienced in molecular dynamics simulations (AMBER), in vitro and in vivo experimentation, pharmaceutical RD, Python programming and Linux-based computational workflows.

Education

Master 2 of Science in Biotechnology & Molecular Engineering Sept. 2026

University of Lorraine, Nancy, France

Relevant coursework: Structural Biology, Protein Engineering, Molecular Modeling, Biophysics, Synthetic Biology, Recombinant Protein Design, Expression and Purification.

Master 1 in Biochemistry, Molecular Biology & Cellular Regulation 2025 – 2026

University of Lorraine, Nancy, France

Relevant coursework: Molecular and Cellular Biology, Experimental Molecular Biology, Biochemistry, Enzymology, Macromolecular Structure and Conformation, Structural Biology, Biomolecular Recognition, Biomolecule Engineering.

Master of Science in Pharmaceutical Biotechnology 2021 – 2023

Higher School of Biotechnology, Algeria

Relevant coursework: Biomedicine Production, Pharmacology and Toxicology, Bioinformatics, AI Applied to Biotechnology, Molecular Modeling and Drug Design, Cell Culture and Innovative Therapies, Proteomics, Bioactive Molecule Synthesis, Natural Product Extraction.

Engineering Degree in Biotechnology 2018 – 2023

Higher School of Biotechnology, Algeria

Multidisciplinary training in molecular biology, microbiology, enzymology, immunology, pharmacology, bioinformatics, biostatistics and biotechnology process development.

Research Experience

Research Intern – Computational Structural Biology Jan. 2026 – Present

Laboratory of Theoretical Physics and Chemistry (LPCT), University of Lorraine, Nancy, France

- Investigating the effects of post-translational histone modifications on nucleosome compaction through all-atom molecular dynamics simulations.

- Prepared biomolecular systems (PDB 1KX5) using TLEaP, including protein and DNA force-field assignment, solvation and ion neutralization.
- Performed molecular dynamics simulations using the AMBER simulation package (PMEMD).
- Analysed simulation trajectories using CPPTRAJ, including RMSD, RMSF, principal component analysis (PCA), distance measurements and residue–DNA contact analysis.
- Developed Python scripts for trajectory analysis and data visualization in a Linux environment.

Research and Development Intern

2024

Institut Pasteur d'Alger, Algeria

- Performed in vitro cell culture experiments and in vivo toxicity studies using murine models.
- Conducted physicochemical, microbiological and immunochemical analyses for quality control of vaccines and biological products.
- Participated in the characterization and evaluation of biological products in a biomedical research environment.

Pharmaceutical R&D Intern

2023

SAIDAL Pharmaceutical Group, Constantine, Algeria

- Conceived and developed a novel topical formulation for the treatment of atopic dermatitis by repurposing the active pharmaceutical ingredient of an oral antihistamine.
- Designed the formulation strategy, selected pharmaceutical excipients and prepared the topical dosage form.
- Performed physicochemical characterization and pharmaceutical quality control of the developed formulation.
- Evaluated the therapeutic efficacy and safety of the formulation using in vivo experimental models.

Selected Projects

Founder and Formulation Scientist – Natural Cosmetic Formulation Project

2024 – present

- Designed and developed a personal line of natural cosmetic products from concept to commercialization.
- Selected active ingredients and pharmaceutical excipients based on formulation requirements and product stability.
- Developed manufacturing protocols and quality control procedures.
- Designed branding strategy, product communication and commercialization.

Core Competencies

Experimental Expertise

Molecular Biology • Recombinant Protein Expression • Protein Engineering • Structural Biology • Cell Culture (in vitro) • In Vivo Experimentation • Microbiology • Pharmaceutical Formulation • Bioactive Molecule Synthesis • Natural Product Extraction • Pharmaceutical Quality Control • Pharmacological Evaluation

Computational Expertise

Molecular Dynamics (AMBER) • Molecular Modeling • Molecular Docking • Structural Bioinformatics • AlphaFold • Trajectory Analysis (RMSD, RMSF, PCA, Contact Distance Analysis) • Python • MATLAB • Linux • PyMol • VMD

Research Interests

Structural Biology • Protein Engineering • Molecular Modeling • Drug Discovery • Therapeutic Peptides • Antibody Engineering • AI for Drug Discovery